

SKILLZENO ELECTRIC VEHICLE INTERNSHIP

TASK 3 – EV CHARGING ANALYSIS

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1. Objective

To study different EV charging levels and technologies, compare their power requirements and applications, calculate approximate charging time for a selected battery pack, and summarize practical charging considerations.

2. Introduction

EV charging transfers electrical energy from the grid or a charging system to the vehicle battery. Charging systems differ mainly in power level, charging method, installation requirements, and charging time.

3. EV Charging Levels

Level	Typical power	Application	Charging characteristic
Level 1 AC	~1–2 kW	Home / standard outlet	Slow; suitable for overnight charging
Level 2 AC	~3.3–22 kW	Home, workplace, public AC	Faster AC charging
DC Fast	~25–350+ kW	Highways / public stations	High-power, shorter charging stops

4. Charging Technologies

AC charging: Grid AC is supplied to the vehicle and converted to DC by the onboard charger. Level 1 and Level 2 are common AC methods.

DC fast charging: The charging station converts AC to DC externally and supplies controlled DC power to the vehicle battery, enabling higher charging power.

Wireless charging: Electrical energy is transferred without a physical cable, generally through inductive coupling between a ground pad and a vehicle receiver.

5. Charging Comparison

Parameter	Level 1	Level 2	DC Fast	Wireless
Power	Low	Medium	High	System dependent
Charging time	Longest	Moderate	Shortest among common options	Depends on power
Installation	Simple outlet	Dedicated circuit	Specialized station	Specialized equipment
Typical use	Home / overnight	Home / workplace	Highway / public	Convenience-focused use

6. Charging-Time Calculation

Selected battery: 60 kWh **Charger:** 7.2 kW AC **Assumed efficiency:** 90%.

Ideal time = $60 \div 7.2 = 8.33$ hours.

Practical approximate time = $60 \div (7.2 \times 0.90) = 9.26$ hours.

Actual time can differ because charging power may reduce at high state of charge, and temperature, battery condition, charger limits, and vehicle control strategy also affect charging.

7. Factors Affecting Charging Time

- Battery capacity and initial state of charge
- Charger power and vehicle maximum accepted power
- Charging efficiency and conversion losses
- Battery temperature and thermal management
- Battery state of health
- Power reduction near high state of charge

8. Safety Considerations

Suitable protection against overcurrent, short circuit, earth leakage, and abnormal conditions is important. Proper earthing, compatible connectors, cable ratings, insulation, and following charger operating instructions support safe charging.

9. Practical Observation

Lower-power AC charging is useful when the vehicle can remain connected for longer periods. Higher-power DC charging is useful when shorter charging stops are required. Charging equipment should match the vehicle and electrical installation ratings.

10. Conclusion

EV charging ranges from low-power AC charging to high-power DC fast charging, while wireless charging provides an alternative energy-transfer method. Battery energy, charger power, and efficiency directly influence approximate charging time.

11. Learning Outcomes

- Understood Level 1, Level 2, and DC fast charging.
- Compared power, applications, and charging characteristics.
- Calculated approximate charging time for a 60 kWh battery.
- Understood practical and safety factors affecting EV charging.